

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA0706

COURSE NAME: COMPUTER NETWORKS

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I __N.Umasri (192424240)_ (Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:N.Umasri

1. CRC (Cyclic Redundancy Check)

Given:

- Data = **11100011**
- Generator Polynomial = $X^3 + 1$
- Generator = **1001**

(i) Determine the transmitted message

Step 1: Append 3 zeros to the data (degree = 3)

11100011 → 11100011000

Step 2: Perform modulo-2 division using divisor **1001**

Remainder = 001

Step 3: Append the remainder to the original data.

Transmitted message = 11100011001

(ii) Leftmost bit is inverted

Received message becomes:

01100011001

The receiver divides the received bits by **1001**.

- **Remainder ≠ 000**
- Hence, the CRC check **fails**.

Answer:

- The receiver gets a **non-zero remainder**.
- Since the remainder is not zero, it detects that an **error has occurred** during transmission.

2(a). Throughput Calculation

Given:

- Bandwidth = **10 Mbps**
- Frames/minute = **12,000**

- Frame size = **10,000 bits**

Formula

Throughput = Frames/sec × Bits/frame

Frames/sec

= $12000 \div 60$

= **200 frames/sec**

Throughput

= 200×10000

= **2,000,000 bits/sec**

= **2 Mbps**

Answer

Throughput = 2 Mbps

2(b). Channel Capacity

Given

- Bandwidth (B) = **3000 Hz**
- Signal-to-noise ratio = **3162**

Formula (Shannon Capacity)

$C = B \log_2(1 + S/N)$

C

= $3000 \times \log_2(3163)$

$\log_2(3163) \approx \mathbf{11.63}$

C

= 3000×11.63

= **34,890 bps**

$\approx \mathbf{34.9 \text{ kbps}}$

Answer

Channel Capacity $\approx 34.9 \text{ kbps}$

3. Hamming (12,8) Code

Analysis

The hospital uses **Hamming (12,8)** error-correcting code.

- Every **8-bit patient data** is converted into a **12-bit codeword**.
- Four parity bits are added.
- If one bit changes because of wireless interference:
 - The receiver checks the parity bits.
 - The incorrect bit position is identified.
 - The receiver automatically corrects the bit.
- Doctors receive the correct patient information without retransmission.

Answer

- Hamming (12,8) detects and corrects **single-bit errors**.
- It ensures reliable transmission of patient monitoring data.
- No retransmission is required, reducing delay.

4(a). IPv4 Subnetting

Given

Network = **172.16.10.0/24**

Subnet Mask = **/27**

Number of subnets

Borrowed bits

$$= 27 - 24$$

$$= 3$$

Subnets

$$= 2^3$$

$$= 8$$

Hosts per subnet

Host bits

$$= 32 - 27$$

$$= 5$$

Hosts

$$= 2^5 - 2$$

$$= 30 \text{ usable hosts}$$

Subnet ranges

Subnet	Range
172.16.10.0	0–31
172.16.10.32	32–63
172.16.10.64	64–95
172.16.10.96	96–127
172.16.10.128	128–159
172.16.10.160	160–191
172.16.10.192	192–223
172.16.10.224	224–255

$$\text{IP} = 172.16.10.78$$

Belongs to

$$172.16.10.64/27$$

Network Address

$$172.16.10.64$$

Broadcast Address

$$172.16.10.95$$

$$\text{IP} = 172.16.10.95$$

Yes.

It is the **broadcast address** of the same subnet.

Answers

- Total subnets = **8**
- Usable hosts = **30**
- Network address = **172.16.10.64**
- Broadcast address = **172.16.10.95**
- **172.16.10.95 belongs to the same subnet but is the broadcast address.**

4(b). IPv6

Given

2001:0db8:0000:0000:0000:ff00:0042:8329

Compressed form

2001:db8::ff00:42:8329

Expanded form

2001:0db8:0000:0000:0000:ff00:0042:8329

Network Prefix (/64)

2001:0db8:0000:0000::/64

Number of Hosts

Host bits

= 128 – 64

= **64**

Hosts

= **2^{64}**

= **18,446,744,073,709,551,616**

Answers

- Compressed Address = **2001:db8::ff00:42:8329**
- Expanded Address = **2001:0db8:0000:0000:0000:ff00:0042:8329**
- Network Prefix = **2001:db8:0:0::/64**
- Total Host Addresses = **2^{64} (18,446,744,073,709,551,616)**

5. Online Examination Scenario

Primary Networking Problem

Flow Control

Explanation

- The server transmits data at **10 Gbps**, while students receive data at lower speeds (100 Mbps, 20 Mbps, or 5–10 Mbps).
- Faster transmission overwhelms slower receivers.
- Receiver buffers become full (**buffer overflow**).
- Packets are dropped.
- Retransmissions occur.
- Acknowledgments are delayed.
- Network congestion increases.
- Students experience incomplete downloads and portal freezing.

Conclusion

Flow control ensures that the sender transmits data only as fast as the receiver can process it. By matching the sender's transmission rate with the receiver's processing capability, flow control prevents buffer overflow, reduces retransmissions, improves network efficiency, and ensures reliable delivery of examination papers.